

1A: P: n is prime T: $n = 2$ O: n is odd

$$P \rightarrow (T \vee O)$$

1B: D: f is differentiable C: f is continuous S: f is smooth

$$D \rightarrow (C \wedge S)$$

1A: P: Being a good programmer C: Having good comms J: Getting good job

$$(P \wedge C) \rightarrow J$$

2: $((p \rightarrow q) \wedge (q \rightarrow r)) \leftrightarrow (p \rightarrow r)$

Not a tautology because not all final values are true, therefore expression is not true for all values.

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$p \rightarrow r$	$((p \rightarrow q) \wedge (q \rightarrow r)) \leftrightarrow (p \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	F	T	F
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	F
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

3: $(p \rightarrow (q \rightarrow r)) = (q \rightarrow (p \rightarrow r))$

$(p \rightarrow (q \rightarrow r))$	RULES	$(q \rightarrow (p \rightarrow r))$	RULES
$\neg p \vee (q \rightarrow r)$	Implication Rule 1	$\neg q \vee (p \rightarrow r)$	Implication Rule 1
$\neg p \vee (\neg q \vee \neg r)$	Implication Rule 1	$\neg q \vee (\neg p \vee r)$	Implication Rule 1
$(\neg p \vee \neg q) \vee r$	Associative	$(\neg q \vee \neg p) \vee r$	Associative
		$(\neg p \vee \neg q) \vee r$	Commutative

$(\neg p \vee \neg q) \vee r = (\neg p \vee \neg q) \vee r$, therefore expressions are logically equivalent. Also in DNF form, showing that it is logically equivalent.

4.

Propositions:

C = Cook telling truth H = Handyman telling truth

G = Gardner telling truth B = Butcher telling truth

1. $B \rightarrow C$
 2. $C \leftrightarrow G$
 3. $H \oplus G$
 4. $\neg H \rightarrow \neg C$
- If the **butcher is telling the truth**, then the cook must also be telling the truth (1). Therefore, the gardener must also be telling the truth and the butcher must be lying (3). But if the butcher is lying, the cook must also be lying too (4). This is a **contradiction** because the cook cannot be lying and telling the truth at the same time. **Therefore, the butcher must be lying.**
 - If the **handyman is lying**, then the gardener must be telling the truth (3). If the gardener is telling the truth, then the cook must also be telling the truth (2). But if the handyman is lying, then the cook must also be lying (4). This is a **contradiction** because the cook cannot be lying and telling the truth. **Therefore, the handyman must be telling the truth.**
 - If the **gardener is lying**, the cook must also be lying (2). The rest of the statements check out as the handyman was proven to be telling the truth and the butcher proved a liar through the above statements. Therefore, this is the only solution to the problem.

Handyman = truth | Cook = lying | Gardner = lying | Butcher = lying

5.

$((p \vee r) \wedge (r \rightarrow q)) \wedge \neg q$

p	q	r	$p \vee r$ (1)	$r \rightarrow q$ (2)	$(1) \wedge (2)$ (3)	$\neg q$	$((p \vee r) \wedge (r \rightarrow q)) \wedge \neg q$
T	T	T	T	T	T	F	F
T	T	F	T	F	F	F	F
T	F	T	T	T	T	T	T
T	F	F	T	T	T	T	T
F	T	T	T	T	T	F	F
F	T	F	F	F	F	F	F
F	F	T	T	T	T	T	T
F	F	F	F	T	F	T	F

CNF FORM: $(p \vee \neg q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge (\neg p \vee \neg q \vee r)$

The CNF form only uses $\{ \vee, \neg, \wedge \}$. Only the rows that flashed true for the statement are included in the CNF form, showing the solutions to the formula.

$\{ \neg, \wedge, \vee, \rightarrow, \leftrightarrow \}$

6.

$(p \oplus q) \vee (q \vee \text{TRUE}) = \text{EXAMPLE FORMULA}$

$(p \oplus q) \vee (q \vee \text{TRUE})$	Proposition Formula
$(p \oplus q) = (p \wedge \neg q) \vee (\neg p \wedge q)$	Logical Equivalence (???)
$(q \vee \text{TRUE}) = \text{TRUE}$	Domination Law
$(p \wedge \neg q) \vee (\neg p \wedge q) \vee \text{TRUE} = (p \wedge \neg q) \vee (\neg p \wedge q)$	Domination Law
$(p \wedge \neg q) \vee (\neg p \wedge q)$	DNF FORM

$\{ \oplus, \vee, \text{TRUE} \}$ is functionally complete because any formula with these three operations can be converted to DNF form. As shown above, any compound proposition that can be converted into DNF is equivalent to any other formula in DNF, therefore proving that the set of connectives is FUNCTIONALLY COMPLETE.